

Surveillance Video-and-Language Understanding: From Small to Large Multimodal Models

Tongtong Yuan¹, Xuange Zhang², Bo Liu³, Kun Liu⁴, Jian Jin⁵, and Zhenzhen Jiao⁶

Abstract—Surveillance videos play a crucial role in public security. However, current tasks related to surveillance videos primarily focus on classifying and localizing anomalous events. Despite achieving notable performance, existing methods are restricted to detecting and classifying predefined events and lack satisfactory semantic understanding. To tackle this challenge, we introduce a novel research avenue focused on Video-and-Language Understanding for surveillance (VALU), and construct the first multimodal surveillance video dataset. We manually annotate the real-world surveillance dataset UCF-Crime with fine-grained event content and timing. Our newly annotated dataset, UCA (UCF-Crime Annotation), contains 23,542 sentences, with an average length of 20 words, and its annotated videos are as long as 110.7 hours. Moreover, we evaluate SOTA models on five multimodal tasks using this newly created dataset, establishing new baselines for surveillance VALU, from small to large models. Our experiments reveal that mainstream models, which perform well on previously public datasets, exhibit poor performance on surveillance video, highlighting new challenges in surveillance VALU. In addition to conducting baseline experiments to compare the performance of existing models, we also propose novel methods for multimodal anomaly detection tasks and finetune multimodal large language model models using our dataset. All the experiments highlight the necessity of constructing this multimodal dataset to advance surveillance AI. Upon the experimental results mentioned above, we conduct further in-depth analysis and discussion. The dataset and codes are provided at <https://xuange923.github.io/Surveillance-Video-Understanding>.

Index Terms—Surveillance video understanding, multimodal learning, dataset annotation, multimodal large language learning.

I. INTRODUCTION

SURVEILLANCE videos are indispensable for public security. The integration of advanced technologies like Artificial Intelligence and Big Data with surveillance systems

has revolutionized public security management [1], [2]. A variety of surveillance-video-oriented tasks have gained extensive research attention in recent years, such as anomaly detection [3], UAV tracking [4], and recognition of unusual or human activities [5], [6]. Involving more human/object activities, these surveillance videos for anomaly detection have attracted our attention. However, current datasets for surveillance anomaly detection [3], [7], [8], [9], [10], [11], [12] typically offer only categorical labels and timing of anomalous events, necessitating the predefinition of all possible categories. As a result, existing methods are confined to the detection and classification of known events, lacking the ability to semantically comprehend the content of the videos. Automatic comprehension of surveillance video content is vital for improving current investigative practices. Surveillance applications often require the ability to search for specific events using queries, rather than broad categories, that is, employing queries to fetch relevant events from surveillance footage. Concurrently, the field of intelligent surveillance is moving towards multimodality, with a particular focus on the interaction between video and text.

Numerous multimodal video datasets [13], [14], [15] have been made publicly available, facilitating the exploration of diverse multimodal learning tasks [16], [17], [18] aimed at achieving a semantic understanding of video content. Despite this, to the best of our knowledge, multimodal learning tailored to surveillance videos remains an underexplored area. For example, some tasks for investigating detailed cases in public security such as temporal sentence grounding in surveillance video, and surveillance video captioning, have been rarely proposed and studied. A primary reason for this gap is the absence of sentence-level language annotations in existing surveillance datasets, which impedes the advancement of learning for these multimodal tasks. Another implicit reason is the increased difficulty in learning multimodal patterns from surveillance videos due to the unique characteristics they possess, which differentiate them from conventional video datasets. Therefore, this calls for a timely multimodal surveillance dataset to validate the challenges of multimodal surveillance video learning and facilitate the development of surveillance AI.

To tackle the aforementioned challenges, we suggest an enhancement of the current surveillance video datasets designed for anomaly detection by incorporating multimodal elements. Specifically, the enriched multimodal surveillance video dataset should encompass authentic video footage, and detailed descriptions of events along with their temporal

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Tongtong Yuan, Xuange Zhang, and Bo Liu are with the School of Computer, Beijing University of Technology (BJUT), Beijing 100124, China (e-mail: yuantt@bjut.edu.cn; szxg923@emails.bjut.edu.cn; liubo@bjut.edu.cn).

Kun Liu is with JD.COM., Beijing 100176, China (e-mail: liu_kun@bupt.cn).

Jian Jin is with China Academy of Information and Communications Technology (CAICT), Beijing 100083, China (e-mail: jin.jian@caict.cn).

Zhenzhen Jiao is with Beijing Teleinfo Technology Company., Ltd., Beijing 100083, China, and also with China Academy of Information and Communications Technology (CAICT), Beijing 100191, China (e-mail: jiaozhenzhen@teleinfo.cn).

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Fig. 1. Annotation examples in our UCA dataset, including fine-grained sentence queries and the corresponding timing.

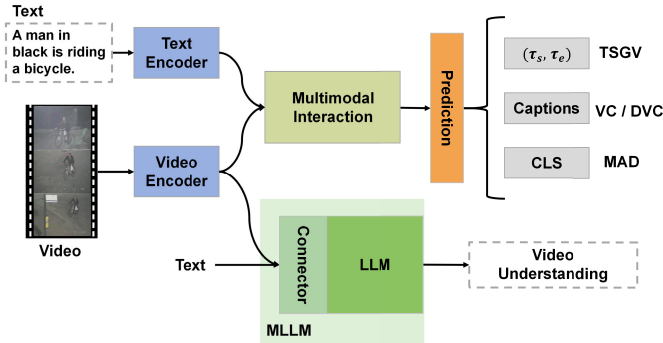


Fig. 2. Integrated multimodal surveillance video understanding framework, including small and large models.

markers. Therefore, we investigated several surveillance datasets [3], [7], [8], [9], [19], [20] and selected UCF-Crime [9] as the foundation of our new dataset. Because UCF-Crime is the largest real-world surveillance dataset and contains a variety of realistic anomalies. To obtain a multimodal surveillance dataset, we contribute new annotations for UCF-Crime [9]. Our dataset is termed UCA (UCF-Crime Annotation), and it is collected by making manually fine-grained annotations of event content and event timing on UCF-Crime [9]. Some examples are shown in Fig. 1.

Compared with UCF-Crime, the main features of our UCA include: (1) The annotation information is relatively fine-grained, and includes as many event descriptions as possible. (2) Considering the realistic demand for temporal localization in the security field, our UCA also includes event timing along with the activity descriptions. (3) UCA can support multiple multimodal understanding tasks, such as Temporal Sentence Grounding in Videos (TSGV) [16], Video Captioning (VC) [17], Dense Video Captioning (DVC) [18], Multimodal Anomaly Detection (MAD) [21], Multimodal Large Language Model (MLLM). To underscore its significance in our work, we have designated MLLM as a separate task branch. Fig. 2 illustrates the comprehensive model architecture that encompasses these tasks.

Our main contributions can be summarized as follows:

- We establish the first real-world and multimodal surveillance dataset with both language descriptions and event timing, which is a comprehensive dataset for the surveillance field to develop the multimodal understandable capacity of machine intelligence.
- We are the first to establish the comprehensive baselines for multimodal tasks (*i.e.*, TSGV, VC, DVC) on surveillance videos. These results provide foundational baselines for understanding surveillance videos. In experiments on MAD, we propose a new method which can improve the

performance of anomaly detection by providing a basic surveillance video captioning model as a plug-and-play module.

- We are the first to test the performance of several MLLMs for VC. Furthermore, we finetune the best-performing MLLM on UCA with LoRA for the first time, significantly enhancing the video description capabilities of the MLLM.
- We thoroughly analyze these experiments from small to large models and have discovered that mainstream models, commonly used in publicly conventional VALU datasets, exhibit poor performance in multimodal surveillance video scenarios. This suggests that the basic learning model architectures must be modified and designed based on the unique characteristics of surveillance video datasets. This finding also underscores the necessity of constructing our dataset and highlights the challenges associated with multimodal surveillance video learning.

The conference version of this work has been presented at CVPR2024 [22]. Main extensions of this paper are as follows: (1) For TSGV, VC, DVC, and MAD tasks, we provide more detailed descriptions of the experimental setup, supplemented with comprehensive visualization results and analysis. (2) We have introduced new experiments with Multimodal Large Language Models (MLLMs) on our dataset. Several commonly used MLLMs were selected and tested for the video captioning task on our dataset. (3) Through these experiments, we have evaluated the performance of each model and conducted a comprehensive comparison of the differences between them. Based on these assessments, we further choose the top-performing MLLM and finetune it to enhance its video description generation capabilities. (4) We have provided more comprehensive background information and comparisons with existing datasets. To help readers better understand the dataset, we further detail the data collection process and annotation guidelines.

The rest of this article is organized as follows. Section II provides an overview of related work. Section III delivers a detailed description of the UCA dataset. Section IV showcases the experimental results of the UCA dataset across various multimodal tasks. Finally, Section V concludes the paper and discusses the impact of our UCA dataset.

II. RELATED WORK

A. Surveillance Video Datasets

Table I presents the comparison of different Surveillance datasets. The majority of surveillance video datasets have some limitations on the number of videos or the degree of reality, such as UCSD Ped1 [3], UCSD Ped2 datasets [3], Avenue dataset [7], Subway dataset [19], ShanghaiTech Campus dataset [8], NWPU [20], etc. In contrast, Sultani et al. [9] have developed a dataset that captures real-world scenarios more authentically, known as UCF-Crime. It comprises 1,900 surveillance videos depicting 13 types of real-world anomalies, including Abuse, Burglary, Explosion, etc. However, UCF-Crime provides annotations solely for anomaly categorization,

TABLE I
COMPARISON OF DIFFERENT SURVEILLANCE DATASETS

Dataset	VideoCount	Abnormal Event Classes	Dataset Length(h)	Total Frames
Subway Entrance [19]	1	5	1.5	86,535
Subway Exit [19]	1	3	1.5	38,940
UCSD Ped1 [3]	70	5	0.08	14,000
UCSD Ped2 [3]	28	5	0.08	4,560
Avenue [7]	37	14	0.5	35,420
ShanghaiTech [8]	437	11		317,398
NWPU [20]	547	28	16	1,466,073
UCF-Crime [9]	1900	13	128	13,769,300

TABLE II
COMPARISON OF THE STATISTICS OF OUR UCA AND OTHER
MULTIMODAL VIDEO DATASETS. **AVG WORD** MEANS THE
AVERAGE NUMBER OF WORDS PER SENTENCE.
TEMP. ANNO. MEANS TEMPORAL ANNOTATION.
THE STATISTICS OF PREVIOUS DATASETS
HAVE BEEN RECORDED IN [24]

Dataset	Domain	#Videos	Duration(h)	#Queries	Avg word	Temp. anno.
MSVD [25]	Open	1,970	5.3	70,028	8.7	✓
TACos [13]	Cooking	127	10.1	18,818	9.0	✓
YouCook [26]	Cooking	88	2.3	3,502	12.6	✗
MPII-MD [14]	Movie	94	73.6	68,375	9.6	✓
ActivityNet Captions [18]	Open	20,000	849.0	73,000	13.5	✓
DiDeMo [27]	Open	10,464	144.2	41,206	7.5	✓
VATEX [28]	Open	41,250	114.6	825,000	15.2	✓
MSRVTT [29]	Open	7,180	41.2	200,000	9.3	✓
MAD [30]	Movie	650	1207.3	384,600	12.7	✓
UCA (Ours)	Surveillance	1,854	121.9	23,542	20.15	✓

which can only be used for abnormal detection. Consequently, within the domain of surveillance video research, there is a dearth of datasets that can support more intricate multimodal learning tasks, such as moment retrieval and video captioning.

To address this issue, we contribute new annotations on the largest surveillance video dataset UCF-Crime. We manually annotated the event content and event occurrence time for 1,854 videos from UCF-Crime, called UCF-Crime Annotation (UCA). Additionally, we have noted that the paper SAVCHOI [23] mentions annotations on the UCF-Crime dataset for monitoring suspicious activities. However, SAVCHOI, which only annotated summaries for 300 videos and lacked detailed temporal information on activities, is not directly comparable with our UCA. The annotations from SAVCHOI are not included in our UCA dataset.

B. Multimodal Video Datasets

Recently, numerous video datasets have been released for different video-language-understanding tasks. We review some video datasets widely utilized in various video-understanding tasks [31], [32], and list the comparison of the statistics of our UCA and other conventional video datasets for multimodal learning in Table II. All of these video datasets consist of high-quality videos and have been annotated with detailed descriptions, which have played indispensable roles in video and language tasks. The main difference between our dataset and other multimodal datasets lies in the difference in domain. Our dataset is specifically designed for the surveillance field. We have provided detailed descriptions of abnormal behaviors in surveillance videos, which are not typically present in general datasets. Additionally, there are also differences in aspects such as video numbers, duration, queries, etc. It is worth noting that our UCA has the longest word length.

C. Multimodal Video Learning

In the following, we briefly review some mainstream multimodal video learning tasks. For more completed reviews, we refer readers to [17], [33], [34], [35], [36], and [37].

- Temporal sentence grounding in videos (TSGV) is a recent multimodal task [16], [33], [38], [39], [40], which learns the temporal activity localization from a given video with respect to a given language query, *i.e.*, the goal of the task is localizing the start and end times for the described activity inside the video. TSGV can be applied in surveillance video to retrieve the event and moment.
- Video captioning (VC) is a multimodal task of producing a natural-language utterance, which aims at describing the visual content of a video. It plays an important technical role in the demand for automatic visual understanding and content summarization [17], [34], [41], which is also important in surveillance video understanding.
- Dense video captioning (DVC) aims to acquire the temporal localization and captioning of all events in an untrimmed video. Surveillance video analysis can also benefit from Dense video captioning. It is more challenging than standard video captioning, which aims to generate a single caption for a video clip [17], [18].
- Multimodal anomaly detection (MAD) is proposed by Chen et al. [21], which fused video caption features with original visual and temporal features for anomaly detection in surveillance videos. But the video caption information is derived from Swinbert [42] trained on open-domain dataset VATEX [28]. We aim to introduce surveillance-domain VC models to generate captions to enhance the multimodal anomaly detection task.
- The Multimodal Large Language Model (MLLM) for VALU aims to enhance the interaction between visual encoders and large language models, enabling a broader range of tasks. The Frozen [43] demonstrated the benefits of using pre-trained language models as decoders in vision-language models. Subsequently, models like Flamingo [44] and BLIP-2 [45] further improved upon this foundation, effectively aligning visual features with language models and training with large-scale image-text datasets, showcasing impressive learning capabilities. Several studies have since leveraged instruction tuning to harness the potential of LLMs in video understanding tasks. MiniGPT-4 [46] directly aligns visual information with the language model by integrating BLIP-2 and Vicuna. VideoChat [47], VideoChatGPT [48], and Valley [49] utilize ChatGPT to generate video instruction-tuning data, with the goal of enhancing instruction-following capabilities. VideoChat2 [50] further strengthens the compatibility between the video encoding module and LLMs through a three-stage training approach. Notably, MLLM for surveillance VALU has not been explored. Therefore, we will conduct a comprehensive comparison of various MLLMs on our UCA, and finetune a specific MLLM for the surveillance domain.

TABLE III
DATA SPLIT AND VIDEO STATISTICS IN OUR UCA

Video Statistics	UCF-Crime	UCA	UCA train	UCA val	UCA test
#Video	1,900	1,854	1,165	379	310
Video length	127.5h	121.9h	75.5h	21.2h	25.2h
Annotated video length	—	110.7h	73.7h	16.4h	20.6h

With the manually fine-grained annotations, our UCA dataset can be used in various multimodal video learning tasks, including but not limited to the above five tasks. We select SOTA methods of these tasks as baselines in the following experiments, by considering various factors, including the novelty, the differences between methods, the stability of experimental results and the openness of the code, etc. These experiments are detailed in **Sec. VI**.

III. THE UCA DATASET

Our dataset is derived from the UCF-Crime dataset. We perform fine-grained language annotations on UCF-Crime, documenting each event or change in the videos with detailed descriptions and timestamps. This annotation process results in a novel dataset named UCA, which is the first large-scale multimodal surveillance video dataset designed for various multimodal tasks. We provide an overview of the UCA dataset, covering data collection, annotation, analysis, comparison with existing datasets, and ethical considerations.

A. Video Collection

All the videos in our UCA dataset are sourced from UCF-Crime [9], a real-world surveillance dataset released at CVPR 2018. The dataset encompasses a wide range of event categories, including Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism, and Normal Videos. During the video collection, we filtered some low-quality videos in the original UCF-Crime to ensure the quality and fairness of our UCA. These low-quality videos encompassed instances of repetition, severe occlusion, and excessively accelerated playback, which affected the clarity of manual annotations and the precision of event timing localization. Therefore, we removed 46 videos from the original UCF-Crime dataset, resulting in 1,854 videos for UCA. The statistics of UCF-Crime and our UCA in terms of the number of videos in different categories are shown in Table IV. The data split in UCA is shown in Table III.

B. Annotations

When labeling a video from UCF-Crime, our goal is to create fine-grained annotations. We record the starting and ending times for each event, with a time interval accuracy of 0.1 seconds. The annotated video length of UCA is 110.7 hours, accounting for 86.8% of the total video duration of UCA. Fig. 1 presents some annotation examples in UCA.

To meet the needs of different researchers and enhance the convenience of data processing, we have uploaded two versions of the annotation files in our dataset project, namely txt

and json formats. The original annotation data was collected in txt format, which includes the corresponding video name, event start time, event end time, and annotation content. This format is simple and easy to read, facilitating preliminary data collection and quick reference. Additionally, considering the needs of research and experimentation, we have converted these txt annotations into json format for ease of use in subsequent experiments, which also includes the original video duration. Specific annotation files can be found on our website.

C. Annotation Guidelines and Agreement Measures

To ensure the accuracy and consistency of the annotations, we provided the annotators with comprehensive annotation instructions. Here are the main annotation guidelines we established prior to commencing the labeling work:

- Fine-grained annotation principle: we emphasize the granularity of the data, considering each event with changes (including changes in human or object states) as an individual data point.
- Time precision: time should be accurately recorded up to 0.1 seconds, allowing for overlaps in the start and end times of multiple samples.
- Rich sentence descriptions: we encourage the use of rich sentence descriptions that enhance semantic understanding, employing techniques such as the use of adjectives to provide detailed coverage of the objects present in the scene.
- Region of Interest (ROI) descriptions: in cases where multiple regions in a single frame experience simultaneous actions or events, annotators may use terms like “top-left corner” or “middle” to differentiate and describe the changes in the states of people or objects within those specific regions.
- Handling intense changes: Annotators address rapid scene changes by merging short-term and long-term descriptions. They initially detail changes in brief intervals to capture minor video variations. These fragmented short-term notes are then synthesized into a single, cohesive narrative.
- Complex environment descriptions: In environments lacking clear primary events or distinct actions, annotators are tasked with providing a general overview to aid in fully understanding the scene context.

To ensure clarity and accuracy in the dataset annotation, we recruited 10 volunteers with computer backgrounds and formed a review team of 3 AI researchers. Before starting, all annotators received comprehensive training. Annotators watched the videos repeatedly to ensure accurate event positioning and description. The review team conducted validation checks every 100 instances to maintain annotation quality, focusing on consistency across annotators. After completing all annotations (23,542 sentence-level queries), the reviewers conducted a final review of the data. The entire process took approximately two months. To ensure agreement among annotators, we imposed specific constraints and guidelines, provided comprehensive annotation guidelines, trained annotators thoroughly, encouraged ongoing communication with reviewers for clarification and feedback, and implemented

TABLE IV
THE COMPARISON OF UCF-CRIME AND UCA IN THE VIDEO NUMBERS OF DIFFERENT CATEGORIES

Video numbers	Abuse	Arrest	Arson	Assault	Burglary	Explosion	Fighting	Road Accidents	Robbery	Shooting	Shoplifting	Stealing	Vandalism	Training Normal	Testing Normal	Summary
UCF-Crime	50	50	50	50	100	50	50	150	150	50	50	100	50	800	150	1900
Our UCA	50	50	50	48	100	50	50	148	149	50	50	100	49	764	146	1854

TABLE V

STATISTICS OF ANNOTATION IN OUR UCA DATASET, INCLUDING THE NUMBER OF QUERIES (#QUERY), THE NUMBER OF WORDS PER QUERY (#WORD/QUERY), AS WELL AS THE NUMBERS OF NOUNS (#NOUNS), VERBS (#VERBS), AND ADJECTIVES (#ADJ)

Annotation Statistics	#Train	#Val	#Test	#Summary
#Query	15,677	3,534	4,331	23,542
#Word/Query	20.45	18.82	20.13	20.15
#Nouns	25,333	5,633	6,748	37,714
#Verbs	26,817	5,569	6,959	39,345
#Adj	12,880	2,963	3,571	19,414

a thorough review and validation process to address any discrepancies or inconsistencies. These measures aimed to ensure the high reliability and quality of the annotated dataset.

D. Dataset Statistical Analysis

As detailed in Table III, our UCA dataset encompasses 1,854 videos, segmented into Train, Validation, and Test subsets based on video length and original categories. UCA's video modality closely mirrors the UCF-Crime dataset. Therefore, we focus on analyzing statistical information on the language modality.

The dataset's annotated video footage totals 110.7 hours as shown in Table III., with Fig. 3 (a) illustrating the distribution of event durations. The majority of events are lasting 5-10 seconds, while few events extend beyond 40 seconds. On average, UCA videos last 236.8 seconds, with event annotations averaging 16.9 seconds each.

Table V outlines the query descriptions and average word count per query, averaging around 20 words. Fig. 3 (b) displays the word count distribution, with the majority of sentences falling into the 10-20 word range, followed by 20-30 words. Vocabulary analysis, as seen in Table V, shows a balanced ratio of nouns, verbs, and adjectives at approximately 2:2:1 across Train, Validation, and Test sets, reflecting a uniform annotation style across various videos.

E. Comparison With Existing Datasets

1) *Difference With Video-Language Datasets*: Table II shows our UCA and other video-language datasets for multimodal learning tasks, highlighting UCA's unique focus on the surveillance domain. This comparison reveals that UCA contains a moderate volume of annotated sentences but excels in the level of detail, with the highest average word count per sentence among similar datasets.

The UCA dataset presents significant differences from traditional video datasets in terms of the quality and complexity, duration of the videos, and redundant Information, making it more challenging for multimodal learning tasks. Its characteristics are mainly reflected in the following aspects:

- **Video Quality and Complexity**: surveillance videos often suffer from uneven image quality, complex backgrounds, and intricate events. For instance, they may include low-resolution, blurry images, or scenes affected by environmental lighting. In contrast, traditional video datasets such as MSVD, YouCook, and TaCos feature higher video quality, simpler backgrounds, and more clearly defined and focused events.
- **Video Duration**: there is a notable variance in the length of videos within the UCF-Crime dataset. The dataset contains 47 videos that are over half an hour long, accounting for 2% of the total number of videos, yet their total duration is 59.56 hours, which is 46% of the total video duration. The longest video spans 9.04 hours, while the shortest is just 3.47 seconds. In comparison, most traditional video datasets tend to have shorter durations.
- **Redundant Information**: in practical applications, surveillance videos are continuously recorded around the clock, containing a large amount of redundant information. This includes prolonged periods of irrelevant content, such as empty scenes or areas without activity, which poses higher demands for data processing and event detection. Differently, traditional video datasets are often edited and processed to retain less redundant information, focusing more on specific events or activities.

2) *Difference With Abnormal Video Datasets*: Our research underscores a pivotal distinction from existing abnormal video datasets [8], [9]: we have undertaken the novel and extensive endeavor of sentence-level annotating events within authentic surveillance contexts. This approach transcends the conventional category-based annotation by offering detailed and precise descriptions of events. As a result, our dataset can be used to conduct various VALU tasks in surveillance scenarios. To our knowledge, the field of multimodal surveillance learning remains largely uncharted, underscoring its substantial practical importance and potential for future exploration.

F. Necessity of UCA Dataset

Intelligent surveillance exhibits a trend toward multimodal directions, especially in video-and-text interaction. Automated surveillance video analysis involves generating textual descriptions used for police incident reports. Additionally, detecting suspicious activities also needs multimodal learning tasks. Practical applications often need to search for specific event queries rather than broad categories, *i.e.*, using queries to retrieve events in surveillance videos. These can demonstrate the necessity of multimodal surveillance learning tasks. However, existing surveillance datasets lack sentence-level annotations, which hinders the research on supervised multimodal surveillance learning and results in a suboptimal semantic understanding of surveillance videos. This also

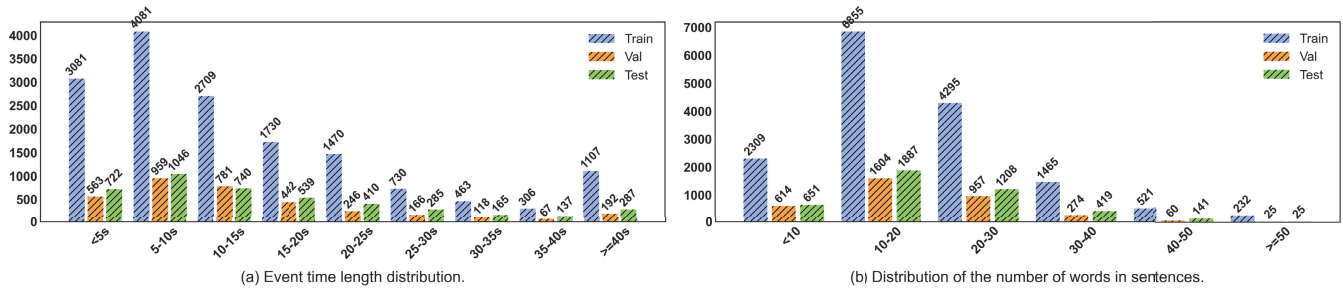


Fig. 3. The duration of annotated events and the number of words in the annotated queries of the UCA dataset.

indicates the necessity of constructing our dataset UCA. Therefore, we build the first multimodal surveillance dataset atop UCF-Crime (the largest real-world surveillance dataset). SOTA models trained on existing event datasets exhibit poor performance in our new dataset, which highlights the necessity of further model research (e.g., the directions of modality alignment, critical information mining shown in Sec IV-F).

G. Ethical Considerations

In ethical considerations for our research, we adhere to stringent ethical standards in the use and annotation of data to avoid any controversy. We emphasize the following information:

- **Data Source:** The videos used in our study are sourced from a publicly available dataset known as UCF-Crime [9]. This dataset does not explicitly mention any privacy or ethical issues. Our work involves sentence-level annotations that describe the events depicted in these videos.
- **Annotation Process:** During our annotation process, we observed that the primary videos did not exhibit any particular focus on race or gender. To further mitigate any potential risks associated with data annotation, we have implemented various measures, such as annotation guidelines and review procedures.
- **Data Usage:** Users should comply with our data licensing agreement. The UCA dataset is released under the Apache License 2.0. Please note that our dataset is intended solely for academic and research purposes. We encourage the academic community and researchers to utilize this dataset to advance the field of surveillance artificial intelligence.

IV. EXPERIMENTS

We conduct experiments on five multimodal tasks using the UCA dataset. For each task, we initially delineate the task itself, its corresponding evaluation metric, and the baseline methods used. Subsequently, we demonstrate the performance of our chosen baselines. We follow the basic experimental settings specified in the respective published papers and public code repositories for these various methods. For MAD, we also propose a new method that involves our pre-trained Swinbert as a textual information branch. For MLLM, we test several benchmarks and further finetune a surveillance MLLM.

A. Temporal Sentence Grounding in Videos

1) **Task:** Temporal sentence grounding in videos (TSGV) aims to identify and localize specific temporal segments within a given video that correspond to a specified language query [33].

2) **Metric:** In existing methods [51], [52], [53], $R@K$ for $\text{IoU}=\theta$ is commonly adopted as the evaluation metric to measure the performance in TSGV. It is defined as the percentage of at least one of the top- K predicted moments that have IoU with ground-truth moment larger than θ [51]. In the following, we set $R@K$ for $\text{IoU}=\theta$ with $K=1, 5$ and $\theta=0.3, 0.5, 0.7$.

3) **Baselines:** We benchmark UCA using 7 different methods (ranging from 2017 to 2023), comparing the novelty, differences, and reproducibility of experiments. CTRL [51] represented a traditional sliding window-based framework of TSGV. SCDM [52] introduced the Semantic Conditioned Dynamic Modulation mechanism, which is a standard anchor-based Method. A2C [54] applied reinforcement learning to this field. For 2D-Map anchor-based methods, 2D-TAN [55] proposed a unique two-dimensional temporal matrix, while MMN [57] further introduced the idea of metric learning. LGI [56] proposed a local-global interaction (LGI) framework as a regression-based method. Momentdiff [58] applied a diffusion model to video moment retrieval.

4) **Implementation Settings:** We utilize the C3D network, pre-trained on Sports1M [59], for visual feature extraction in our experiments. Given its widespread use across models for various datasets, we adopt the C3D network for the UCA dataset to streamline experimentation and ensure straightforward comparability with other studies.

For CTRL [51], we set the sliding window size to 128 or 240 frames (5 or 10 seconds) with a 0.5 overlap and 1 context clip. For SCDM [52], the input video length is set to 512 frames. The settings for the temporal convolutional layers are five temporal dimensions $\{128, 64, 32, 16, 8\}$. For A2C [54], the maximum frame length T_{max} is 10. For LGI [56], the model uniformly samples 128 segments from each video, with the maximum text length set to 50. For 2D-TAN [55] and MMN [57], the number of sampled clips N in visual features is set to 16. For MomentDiff [58], the maximum length of visual features is set to 1,000.

5) **Results and Analysis:** The experimental results from Table VI highlight the significant challenges of the TSGV task in surveillance videos. The $R@1$, $\text{IoU}=0.3$ metrics

TABLE VI
BENCHMARKING OF TSGV BASELINES ON OUR UCA DATASET

Method	IoU=0.3		IoU=0.5		IoU=0.7		FLOPs(G)	Inference time(s)
	R@1	R@5	R@1	R@5	R@1	R@5		
CTRL [51]	6.35	20.09	3.00	10.88	0.74	3.32	0.02	3.03
SCDM [52]	6.49	10.72	3.65	7.39	1.66	4.78	153.41	2.00
A2C [54]	4.25	-	1.78	-	0.44	-	0.02	0.08
2D-TAN [55]	7.76	19.67	4.16	11.54	2.22	6.00	52.42	0.15
LGI [56]	7.71	-	3.26	-	1.18	-	6.02	0.13
MMN [57]	8.68	21.38	4.66	11.98	2.12	5.89	59.35	0.07
MomentDiff [58]	8.15	-	4.46	-	1.71	-	2.68	0.12

Sentence query: The man with the knife dragged the beaten man out of the yard, and the woman in black ran out of the screen.



Fig. 4. Visualization results of TSGV.

are generally below 10%, indicating the difficulty of precise event localization. UCA videos have varying durations, but existing models struggle with long-term temporal information, affecting the fusion of text and video features. Among the numerous benchmarking methods, the A2C model, which employs reinforcement learning mechanisms, exhibits relatively inferior performance on the UCA dataset. However, two models employing 2D Temporal Adjacent Networks, namely 2D-TAN and MMN, demonstrate superior overall performance compared to other methods. This advantage can be attributed to their capability to perceive richer contextual information from video data, thereby better comprehending and capturing these temporal characteristics. Table VI also summarizes the computational complexity and the average inference time per data point for each model. The CTRL model, despite its simple structure, requires considerable time to match video and language features. The SCDM model, with its complex dynamic modulation mechanism, has high computational complexity. The A2C model matches and localizes video and language descriptions through observation and policy networks, effectively controlling both computational load and inference time. Both 2D-TAN and MMN require high FLOPs due to computations across the 2D feature map, but MMN reduces computation costs by sharing moment features within videos. The LGI model maintains low complexity and inference time through local and global video-text interactions. The MomentDiff model uses a generative approach for efficient inference with low computational complexity. Fig. 4 showcases instances where the IoU is greater than 0.7 between predicted and ground truth results in the TSGV task.

B. Video Captioning

1) *Task*: Video captioning (VC) aims to understand a video clip and describe it with language [17].

2) *Metric*: We use the metrics as in [42] and [60], including Bilingual Evaluation Understudy (BLEU) [B@n, n=1,2,3,4] [61], Metric for Evaluation of Translation with Explicit Ordering (METEOR) [M] [62], Recall Oriented Understudy of Gisting Evaluation (ROUGE-L) [R] [63],

and Consensus-based Image Description Evaluation (CIDEr) [C] [64].

3) *Baselines*: We select 6 methods (ranging from 2015 to 2023) for video captioning as our baselines. These methods have various motivations for learning VC: S2VT [65] is a classic work that first proposed an end-to-end model for generating video descriptions, providing important insights for subsequent research. RecNet [66] introduced the encoder-decoder-reconstructor architecture, using the idea of forward and backward flows for video caption generation. MARN [67] improved the traditional encoder-decoder framework and proposed a Memory-Attended Recurrent Network. SGN [60] emphasized the importance of decoded captions and used semantic groups as information units to describe videos. SwinBERT [42] is the first end-to-end video caption generation model fully based on Transformer. CoCap [68] introduced an end-to-end video captioning approach based on compressed videos, achieving faster runtime speeds.

4) *Implementation Settings*: We utilize the pre-trained models library for extracting visual features.

For S2VT [65], we uniformly sample 80 frames from each video, employing pre-trained Inception V4 [69] and VGG16 [70] as the visual encoders. For RecNet [66], the visual encoder also utilizes Inception V4, sampling 80 frames uniformly from each video. The maximum sentence length is limited to 50. For MARN [67], we similarly sample 80 frames from each video, with pre-trained Inception V4 and ResNext101 as visual encoders. The maximum sentence length is set to 50. Initially, 100 epochs of training are conducted on the Attention-based Recurrent Decoder, followed by the integration of the Attended Memory Decoder. For SGN [60], 50 frames are uniformly sampled from each video, using pre-trained ResNet [71] 101 and ResNext101 [72] as visual encoders to extract 2D and 3D features. The maximum sentence length is set to 30, with word embedding sequences obtained through GloVe [73]. The model's similarity threshold τ is set to 0.2, and the coefficient λ for Contrastive Attention loss is 0.16. For SwinBERT [42], each video is segmented into 32 frames, using end-to-end training. VidSwin [74] is initialized with pre-trained weights from Kinetics-600 [75], and the multimodal transformer encoder is randomly initialized. For CoCap [68], before training and testing, the video's minimum edge size is adjusted to 240, and it is compressed using H.264 encoding. The visual encoder is initialized with pre-trained weights from CLIP [76], while the other encoders and the multimodal decoder are randomly initialized.

5) *Results and Analysis*: The performance of video captioning baselines is presented in Table VII. Among these benchmarks, SGN has achieved a higher accuracy in phrase matching due to its semantic group design, effectively integrating annotation information with video frames. Two Transformer-based models, SwinBERT and CoCap, have demonstrated superior performance across multiple metrics, particularly in this CIDEr metric. This substantiates their capability to capture key textual information more effectively. Fig. 5 depicts a prediction example for VC. Overall, surveillance videos have distinctive characteristics, such as blurry

TABLE VII
BENCHMARKING OF VC BASELINES ON OUR UCA. B, M, R, AND C
REPRESENT BLEU, METEOR, ROUGE-L,
AND CIDER, RESPECTIVELY

Method	Features	B1	B2	B3	B4	M	R	C
S2VT [65]	Inception V4	28.27	16.34	9.62	5.56	10.35	25.38	15.44
S2VT [65]	VGG 16	23.53	13.17	7.72	4.53	10.39	24.09	13.92
RecNet_global [66]	Inception V4	25.81	15.06	9.14	5.78	10.42	25.12	16.90
RecNet_local [66]	Inception V4	26.89	15.73	9.52	6.00	10.55	25.52	16.70
MARN [67]	Inception V4 ResNext101	26.16	15.61	10.22	6.63	9.67	23.85	15.33
SGN [60]	ResNet101 ResNext101	29.17	16.93	10.32	6.28	11.73	26.25	18.95
SwinBERT [42]	VidSwin	25.02	15.60	9.98	6.33	11.15	27.16	25.29
CoCap [68]	CLIP	28.53	17.34	10.75	6.57	11.43	28.14	21.24



Ground-Truth: The woman in pink clothes took several things from the bed, when she raised her feet, the baby grabbed her skirt and then let go.

Prediction: The woman in the pink top walked to the side of the bed then stood up and walked.

Fig. 5. Visualization results for VC.

and low-resolution visuals, which will pose more challenges for VC models to learn and align the visual and text modality.

C. Dense Video Captioning

1) *Task:* Dense video captioning (DVC) aims to generate temporal localization and captions for dense events in an untrimmed video.

2) *Metric:* We have assessed the performance from two perspectives. For localization performance, we employed the evaluation tools provided by the ActivityNet Captions Challenge 2018 and used the common metrics [77], [79], like different IoU thresholds (0.3, 0.5, 0.7, 0.9), classic caption evaluation metrics: BLEU, METEOR, CIDEr, and the performance in describing video stories: SODA_c score.

3) *Baselines:* We select three methods (ranging from 2018 to 2022) for DVC with different input features (to increase the diversity of comparisons) as our baselines. These methods have distinct designs and are reproducible. TDA-CG [77] proposed a bidirectional proposal method that effectively utilized past and future context for proposal prediction. PDVC [78] enhanced the representation of event queries and input the localization head and caption head in parallel. UEDVC [79] fully utilized the interdependence between events to detect more diverse and consistent events.

4) *Implementation Settings:* The C3D features used remain consistent with those used in TSGV. The I3D features are directly obtained from an external GitHub repository,¹ which was established by other researchers. For TDA-CG [77], each feature sequence corresponds to 64 frames of content in the video. The maximum sentence length is limited to 30. During training, the Adam optimizer is used. For PDVC [78], we set the number of event queries to 100, and conduct experiments

¹<https://github.com/tianyu0207/RTFM>

TABLE VIII
EVENT LOCALIZATION OF DVC ON UCA

Method	Features	Recall				Precision			
		0.3	0.5	0.7	0.9	0.3	0.5	0.7	0.9
TDA-CG [77]	C3D	32.54	16.64	8.40	2.49	58.79	27.16	8.95	1.44
PDVC [78]	C3D	48.03	28.48	11.80	2.50	67.33	36.19	12.20	1.94
TDA-CG [77]	I3D	31.72	17.09	8.97	2.89	57.88	29.69	12.45	1.91
PDVC [78]	I3D	49.53	28.59	12.56	2.11	68.84	35.36	12.84	1.73

TABLE IX
DVC ON UCA WITH PREDICTED PROPOSALS

Method	Features	Predicted proposals						SODA_c
		B1	B2	B3	B4	M	C	
TDA-CG [77]	C3D	5.50	2.23	0.84	0.38	2.72	4.44	0.98
PDVC [78]	C3D	7.89	3.93	1.65	0.55	3.78	8.58	2.07
TDA-CG [77]	I3D	6.35	2.84	1.22	0.37	3.12	5.51	0.98
PDVC [78]	I3D	8.02	4.22	2.00	0.71	4.06	8.78	2.21

TABLE X
DVC ON UCA WITH GROUND-TRUTH PROPOSALS

Method	Features	Ground-Truth proposals					
		B1	B2	B3	B4	M	C
PDVC [78]	C3D	23.09	11.63	5.38	1.95	10.27	19.16
UEDVC [79]	C3D	28.91	16.54	9.58	5.49	10.59	14.64
PDVC [78]	I3D	23.64	12.58	5.90	2.52	10.56	22.65
UEDVC [79]	I3D	26.00	15.47	9.53	5.71	10.62	18.83

using the standard PDVC model with the LSTMDSA captioneer. The LSTM hidden layer dimension in the caption head is set to 512. The Adam optimizer is used during training. For UEDVC [79], the maximum length of video frames is set to 200, and the maximum sentence length is 50. The number of layers in the independent encoder is set to 1, while the cross encoder has 4 layers. This model's training also employs the Adam optimizer.

5) *Results and Analysis:* The localization performance of the model on the UCA dataset is illustrated in Table VIII. Overall, the model exhibits a tendency for relatively higher recall and relatively lower precision. Across different IoU thresholds, the PDVC method consistently outperforms the TDA-CG method, implying the PDVC method's notable competence in target localization. Dense captioning performance is shown in Table IX and Table X. Using predicted proposals, PDVC surpasses TDA-CG across most metrics, especially in BLEU and METEOR scores, implying superior n-gram matching and semantic alignment in PDVC-generated captions. However, all methods exhibit suboptimal CIDEr performance, suggesting room for enhancing caption diversity. In terms of the SODA_c metric, PDVC's captions better capture the overall video story to some extent compared to TDA-CG. With ground-truth proposals, PDVC outperforms UEDVC in the CIDEr indicator, suggesting a better similarity between its predictions and the ground truth. Fig. 6 displays a prediction example in DVC. DVC tasks in UCA encounter challenges from both TSGV and VC, including learning comprehensive temporal information, retaining crucial video event features, and exploring text information.

D. Multimodal Anomaly Detection

1) *Task:* Anomaly detection is one of the important learning tasks in video surveillance. Previous studies have predominantly focused on spatial and temporal features, however,



Fig. 6. Visualization results for DVC.

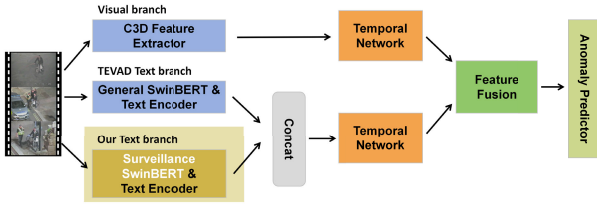


Fig. 7. Our proposed framework for MAD.

in many complex real-world surveillance videos, these visual features are insufficient in capturing rich semantic meanings. Therefore, Chen et al. [21] proposed the learning task of multimodal anomaly detection (MAD), which utilizes captions generated by a video captioning model (*i.e.*, Swinbert [42]) as a text feature source to improve the performance of traditional anomaly detection.

2) *Metric*: As in TEVAD [21], we also employ the micro-averaged AUC (Area Under the ROC curve) by concatenating all frames of the videos to obtain the AUC scores.

3) *Baselines and Ours*: We choose TEVAD [21] as the baselines of MAD, which fused multimodal features, including visual features, temporal features, and caption features. We also propose a new framework for anomaly detection with captions on the basis of TEVAD, which is denoted as enhanced TEVAD, and the framework is shown in Fig. 7. TEVAD [21] deploys Swinbert [42] pre-trained on VATEX [28] (General Swinbert in Fig. 7) as the video caption extractors. VATEX is a large-scale and general video dataset, which provides a general video captioning capacity. Differently, our pre-trained Swinbert on UCA can provide more capacity for captioning the surveillance videos. On the basis of TEVAD, our enhanced TEVAD further introduces our pre-trained Swinbert on UCA (Surveillance Swinbert in Fig. 7, as a plug-and-play module) as a domain-specific feature branch for anomaly detection.

4) *Implementation Settings*: We use the original experiment settings in TEVAD, and deploy feature concat to fuse multimodal features. The main difference between TEVAD and our enhanced TEVAD is that we introduce a new Surveillance Swinbert trained on UCA.

5) *Results and Analysis*: The results in Table XI show our enhanced TEVAD outperforms others. It indicates that our pre-trained Swinbert on UCA can improve anomaly detection accuracy by providing surveillance-domain-specific feature information. In the example of Fig. 8, we can find that our Surveillance Swinbert can generate captions with anomalous descriptions, which is useful in multimodal anomaly detection.

TABLE XI

MULTIMODAL ANOMALY DETECTION (MAD) ON UCF-CRIME, WHERE “VISUAL” DENOTES VISUAL FEATURES

method	visual	General Swinbert	Surveillance Swinbert	AUC
Visual	✓	✗	✗	83.1%
TEVAD [21] (2023)	✓	✓	✗	84.9%
Enhanced TEVAD (Ours)	✓	✓	✓	85.3%



Fig. 8. Examples of different video captioning results in MAD.

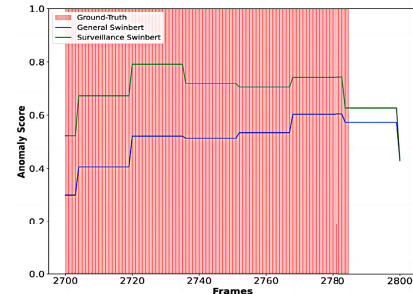


Fig. 9. Comparison of MAD using different text branches. The magnitude of curves represents the anomaly score.

Fig. 9 illustrates the impact of different text branches on anomaly detection result scores in MAD, which extracts frames 2,700-2,800 from the video in UCF-Crime, with the red region indicating frames labeled as anomalies in the ground truth. The two curves represent the results of different text branches obtained using General SwinBERT and Surveillance SwinBERT. Clearly, when using Surveillance SwinBERT to generate descriptive statements, anomalous video frames obtain higher anomaly scores, thereby improving the accuracy of anomaly detection. This also demonstrates the effectiveness of our UCA and our trained multimodal models, which can play an important role in refining existing surveillance tasks.

E. Multimodal Large Language Models

Firstly, we test the performance of several MLLMs for video captioning. Then, we finetune the best-performing MLLM (VideoChat2) on UCA with LoRA, significantly enhancing the video description capabilities of the MLLM.

Task: The multimodal large language model aims to effectively align the visual information of videos with the textual content of LLMs, achieving a deep understanding of video content. In this study, we focus on evaluating the performance of different models in video content understanding, which is similar to Video Captioning tasks. This task requires the model

to accurately describe the specific events occurring in video segments.

Metric: We use BLEU, METEOR, ROUGE, and CIDEr to evaluate the quality of the generated video descriptions.

1) *Baselines of MLLM for Video Captioning:* We selected five different VALU methods as baselines, with code provided by the Shanghai AI Lab.² The Video Chat with StableLM model uses the pre-trained GRiT [80] for keyframe analysis in videos and Tag2Text [81] to generate tags and content descriptions for each frame. Finally, it integrates the information from these different branches into StableLM to produce detailed video descriptions. The Video Chat with MOSS [82] model follows a similar process as StableLM, using Tag2Text and GRiT to generate captions, which are then input into MOSS to answer questions about the video content. MiniGPT-4 [46] aligns visual information from a pre-trained visual encoder with the large language model Vicuna to generate video content descriptions. VideoChat [47] is an end-to-end chat-centric video understanding system that can handle both text and video data simultaneously. VideoChat2 [50] employs a progressive multimodal training paradigm, significantly surpassing other models in video understanding capability.

a) *Implementation settings:* For Video Chat with StableLM, we first perform interval sampling on video clips, sampling up to 30 frames per clip and obtaining a keyframe every 5 frames. The sampled video frames are processed using the Tag2Text model, which utilizes the Swin-Base as its backbone, to generate captioning and tags for each frame. Keyframes undergo dense captioning using content obtained from a pre-trained GRiT model based on the ViT-B version. Finally, the video tags, secondary descriptions, and dense captions are integrated to form a prompt, which is then fed into the 7B parameter StableLM-alpha to produce the final video content description. For Video Chat with MOSS, the process of obtaining video tags, video descriptions, and dense video captions mirrors that of StableLM. The video content description and question information are input into MOSS to generate the video content description. The MOSS weights use a 4-bit quantized version of the *moss-moon-003-sft* model, which allows for inference with an approximate 12GB memory usage. For MiniGPT-4, 4 frames are sampled per video as the default settings. The visual representation is obtained using a model architecture based on BLIP-2, and subsequently, the visual representation and question are input into the 7 billion parameter Vicuna. For VideoChat, during video loading, the maximum sampling frame rate does not exceed 30. The ViT model is used to extract video features, with QFormer aligning these features to the language space as a connector. The pre-trained 7B parameter VideoChat model generates descriptions for each video. For VideoChat2, we test the model using the provided weights, with a maximum frame sampling rate of 30 for videos. The LLM is based on Vicuna-7B.

Considering the UCA dataset only provides descriptions of video clips rather than a question-and-answer format, suitable prompts must be designed for generating video content understanding via MLLM. During the annotation of the UCA

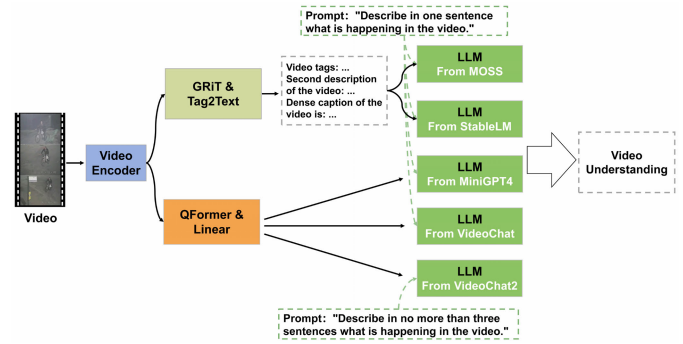


Fig. 10. Integrated Framework for Multimodal Large Language Model Video Captioning Evaluation.

TABLE XII

BENCHMARKING OF MLLM BASELINES ON OUR UCA FOR VC TASK. B, M, R, AND C REPRESENT BLEU, METEOR, ROUGE-L, AND CIDEr, RESPECTIVELY

Method	B1	B2	B3	B4	M	R	C
StableLM	14.36	6.34	2.82	1.30	9.54	17.53	4.59
MOSS	4.72	0.85	0.14	0.00	2.92	4.86	1.57
MiniGPT4	15.83	6.85	2.94	1.38	8.60	18.21	6.77
VideoChat	17.14	6.63	2.52	1.03	8.59	17.06	5.60
VideoChat2	24.08	10.25	4.02	1.82	10.29	20.68	10.07
VC2_Fintune (Ours)	29.73	18.42	11.46	7.08	12.48	27.90	26.77

dataset, annotators were required to follow the principle of “describing the content of the video in natural language.” We aim for the model-generated content to be as close to the real annotations as possible in both style and content. Therefore, for VideoChat2, we use the prompt “Describe in no more than three sentences what is happening in the video.” For other models, we observed that requesting the model to describe in three sentences resulted in a lot of repetitive content, affecting the final test results. Hence, we further restricted the output length with the prompt “Describe in one sentence what is happening in the video.” Fig. 10 shows a visual representation of the comprehensive framework used for evaluation with various large models.

b) *Results and analysis:* Table XII presents the test results of various models on the UCA dataset for the video description task, as well as the performance improvements of the fine-tuned VideoChat2 across all evaluation metrics. Fig. 11 offers a visual representation of the captions generated by the models. It is evident from the image that the scenarios described by MOSS are largely unrelated to the actual video content. StableLM and MiniGPT4 include many incorrect details. VideoChat describes some dynamic events but misses key incidents. VideoChat2 performs the best, generating the most relevant descriptions.

We further analyze the performance of different models. For StableLM, the visual results suggest that its style tends to list objects present in the frame rather than narrating the activities within the video, leading to less satisfactory performance in overall similarity. For MOSS, despite using the same models as StableLM in the first stage of video content analysis, meaning both models input identical information such as image caption, dense caption, and video tag, the quality of the descriptions produced by MOSS is noticeably inferior to that of StableLM.

²<https://github.com/OpenGVLab/Ask-Anything>

		
Groundtruth: The man threw a fire into the car and left and the car started to burn.	Groundtruth: The motorcycle caught fire while refueling people around dispersed and the man in white tried to put out the fire.	Groundtruth: An explosion occurred in a distant building and produced smoke and the glass of the building next to it was shaken.
MOSS: A group consists men carrying suitcases while walking down someones yard at night.	MOSS: A woman walks through different locations while carrying her belongings including children who have bicycles attached under each arm.	MOSS: A group watches from across town while their friend throws his frisbee downrange at some passing vehicles.
StableLM: The video shows a women looking through a windshield of a car as it is parked next to a man, a suitcase standing on the hood of the car, and a man with a hood and suitcase on the street as they are parked next to each other.	StableLM: A child's red motorcycle, a large white pole, a person on a bike, a person walking on the street, a car in the street, a scooter parked by the road, a person on a bike, a car is parked, and a person walking with a large white pole.	StableLM: The main focus of the video is a parked car, a red car, and a graffiti-filled street in a city.
MiniGPT4: The video shows a man standing in front of a car with the engine running.	MiniGPT4: A person is standing outside of a gas station with their arm outstretched in front of a person standing next to a gas pump on the sidewalk in front of a building with windows on the top and bottom.	MiniGPT4: There are several cars parked on the sidewalk in front of a building, with a fire shooting out of the top of the building.
VideoChat: A person is walking in the road and walking into the car, then the car starts to beep continuously and then there are flashes of lights and smoke in the road, and a person is standing on the sidewalk and watching the situation.	VideoChat: The video shows people riding bicycles, pushing motorcycles, and walking in a gas station.	VideoChat: The video shows a small white car driving through a city street, passing by parking spots, parked cars, a fire hydrant, a blue car, and other vehicles.
VideoChat2: A man is seen pouring water on a car. He is wearing a white suit and appears to be cleaning the car.	VideoChat2: A man is seen riding a motorcycle and then getting off of it. He then proceeds to walk away from the motorcycle.	VideoChat2: The video shows a car driving down a street and then stopping. The car is parked on the side of the road.

Fig. 11. Visualization results of MLLM.

This indicates that MOSS has deficiencies in integrating contextual information and outputting coherent descriptions compared to other models. MiniGPT-4 has similar issues to StableLM, with low accuracy in annotations. VideoChat needs improvement in recognizing activities within videos. Among all tested models, VideoChat2 shows a significant advantage, particularly on the CIDEr metric. VideoChat2 identifies the visual content in the videos more accurately in generating structurally sound and semantically coherent captions.

In comparisons of the large models, we find the impact of the large language model component on overall content understanding from StableLM and MOSS. Meanwhile, VideoChat and VideoChat2 highlight the importance of aligning the visual component with the LLM. Thus, how to extract more effective visual information and effectively align it with LLMs would be key factors in improving model performance when integrating text-targeted tasks with other tasks.

2) *Our Fine-Tuned MLLM Based on VideoChat2:* We further finetune VideoChat2 with our UCA dataset. Fig. 12 illustrates the schematic of our model fine-tuning process. In this process, videos are initially encoded through the Vision Transformer, and then the data is passed into the QFormer. The QFormer serves as an intermediary connector, responsible for aligning visual information with the Large Language Model. Subsequently, the visual information is merged with textual information. During the training phase, we input manually annotated video segment descriptions into the model as instructions. In the testing phase, the language model is required to generate video captions based on the posed questions. After the fusion of visual and textual information, it is further fed into LLM, with this segment being fine-tuned using the LoRA technique to enhance the model's comprehension and response to the video content and associated queries.

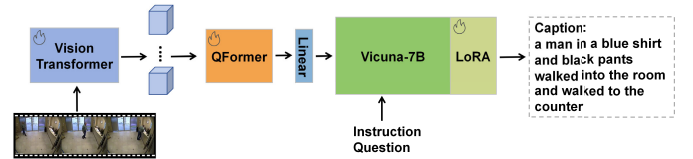


Fig. 12. Our finetuning process for VideoChat2.

a) *Experiment details:* We further fine-tune the VideoChat2 model on the UCA dataset to adapt it to the characteristics of surveillance videos. The LLM is fine-tuned using LoRA, with the parameter rank set to 16 and the scaling parameter, denoted as α , set to 32. The connector undergoes full parameter updating, while the video encoder will be also updated. To keep the generality of the model, we continue to load the original stage 3 weights during training. We sample 8 frames per video.

b) *Results and analysis:* By further exploiting our UCA dataset, we finetune VideoChat2 to obtain better VALU ability. The last row in Table XII details the performance data of our fine-tuned VideoChat2 model, while Fig. 13 visually compares the model's outputs using both the original and fine-tuned weights. It is evident from the table that the fine-tuned model shows a significant increase in all evaluation metrics, indicating that the fine-tuning process substantially enhanced the model's accuracy and comprehensiveness in generating descriptions. The visualizations also reveal that the fine-tuned model can more precisely capture and articulate key events and details within videos.

Moreover, when compared with the mainstream models in traditional video captioning tasks as depicted in Table VII, the video descriptions generated by our finetuned VideoChat2 are



Fig. 13. Comparisons of VideoChat2 and Finetuned VC2.

also markedly superior in quality, with more complete and fluid semantic expression. This suggests that the approach combining LLM with visual models is more effective in understanding video content than conventional methods and can address a broader range of application scenarios. In future work, how to design the visual information processing module and achieve precise alignment with language models is a direction worth exploring in the field of multimodal learning.

F. Discussion

We have conducted experiments on mainstream multimodal video understanding tasks. It is the first time that these tasks are oriented to surveillance video. By analyzing the experiment results, we give the following discussions.

- SOTA methods oriented to conventional videos cannot perform well on our new UCA, which demonstrates the challenges of surveillance VALU. Several instances of these performance gaps lie in multimodal learning between UCA and conventional video datasets are shown in Fig. 14. For TSGV tasks, surveillance videos have long video lengths, leading to the difficulties of temporal grounding. For VC and DVC tasks, due to different domains of videos having particularities, a general model cannot well adapt to a surveillance domain. Moreover, we also found that multimodal models trained from general videos could not capture the abnormal behaviors and generate suboptimal descriptions missing critical abnormal information as shown in Fig. 8.
- For the multimodal anomaly detection task, we can find that surveillance video captioning can play an auxiliary role in anomaly detection by fusing the caption features. The model trained on our UCA can complement the specific capacity of surveillance video understanding, compared with the model trained on conventional videos. This highlights the effectiveness of our UCA. Therefore, we argue that a well-trained multimodal surveillance video learning model can become a promising factor in lifting existing surveillance and security tasks.
- During the testing of multimodal large models, it becomes evident that LLMs offer significant advantages. These models show extensive linguistic knowledge and formidable generative capabilities, supplying richer contextual information and more refined semantic comprehension. This stands in contrast to conventional video captioning models, which often rely on vocabularies derived from datasets, thus limiting their adaptability in open-ended scenarios to an extent. However, LLMs

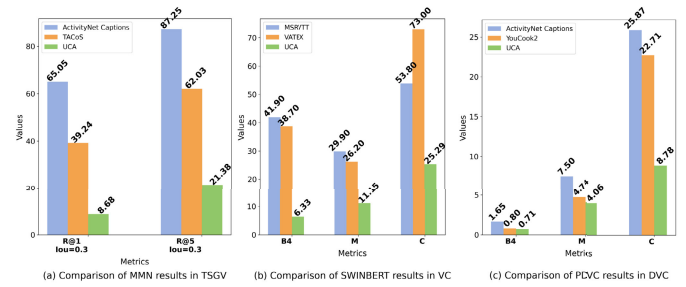


Fig. 14. Illustrations of the performance gaps between our UCA and other conventional datasets.

encounter challenges when processing lengthy sequences of data, and their training and deployment demand considerable computational resources. Therefore, in the process of merging the language modality with other modalities, procuring more valuable visual information and enhancing its alignment with the language modality constitutes a viable strategy for performance improvement. Moreover, finetuning certain parts of the LLMs is an effective approach to resource conservation.

In addition, we have some suggestions for designing a more suitable visual encoder for surveillance VALU tasks:

- Long videos often contain a plethora of redundant information. The commonly used method of uniform sampling based on video duration does not guarantee the capture of critical information, especially in surveillance VALU tasks, where abnormal frames might be missed, leading to a misunderstanding of the video content. Therefore, we suggest designing an appropriate sampling module to recognize critical frames before processing video features.
- Surveillance videos typically have the characteristics of low resolution and a fixed camera perspective, which makes it beneficial to incorporate a module for multi-scale feature extraction and background modeling. Multi-scale feature extraction allows for a better understanding of both the coarse and fine-grained information in the data, which can release the challenges posed by the distant cameras; background modeling assists in distinguishing between abnormal events in the foreground and normal background changes, thereby enhancing the model's ability to handle complex scenes.

However, obtaining such a well-trained surveillance learning model needs the support of datasets. Because the collection of surveillance video data is still challenging, our UCA based on UCF-Crime, also suffers from insufficient data volumes. To solve the limitations of our UCA in data scale, we investigate some existing video datasets and find that TRECVID-VIRAT [83] and MEVA [84] can help expand our multimodal surveillance video dataset. Using MLLMs to auto-generate sentence-level annotations for those surveillance videos is also the key focus of our future work. There are also some interesting research directions:

- independently constructing a series of datasets with new annotations and exploring the ability to transfer our trained models to different datasets.

- sequentially incorporating the newly annotated datasets into the model learning process, and devising a continual learning framework to address data from diverse sources.

V. CONCLUSION AND IMPACT

In the current AI domain, there is a notable lack of research on multimodal surveillance video datasets, despite their potential to enhance social security and daily life. To bridge this gap, we introduce the real-world multimodal UCA dataset, which is developed through the re-annotation of the UCF-Crime dataset. Our annotation strategy prioritizes the delivery of comprehensive event narratives and precise temporal boundaries for events, whether normal or abnormal, occurring within surveillance footage. This effort culminates in a dataset enriched with 23,542 sentence-level descriptions and frame-level time records. Additionally, we conduct experiments on five multimodal tasks with UCA, evaluating the performance of around 20 benchmark methods. These results have demonstrated considerable challenges in the field, underscoring the need for further research in both datasets and models.

Our UCA is the first multimodal surveillance video dataset and can be applied to the research on intelligent public security. In particular, it can provide a fundamental dataset for tasks related to multimodal surveillance video comprehension, which is a novel area within the surveillance domain. Based on UCA, researchers can explore retrieving detailed event queries with temporal information, captioning surveillance videos, and multimodal anomaly detection, to improve technical investigative capabilities. We also provide some practices to develop a surveillance multimodal large language model, which also brings inspiration for the intelligent security community.

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Kun Liu received the B.E. degree in computer science from Yanshan University and the Ph.D. degree from Beijing University of Posts and Telecommunications, Beijing, China. He is currently a Researcher with JD, China. His research interests include multimedia content analysis and computer vision.



Tongtong Yuan received the bachelor's and Ph.D. degrees from Beijing University of Posts and Telecommunications (BUPT), Beijing, China. She is currently a Faculty Member with the School of Computer Science and Technology, Beijing University of Technology, Beijing. Her research interests include computer vision and intelligent surveillance.



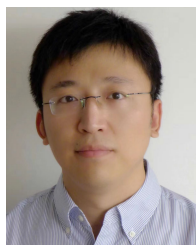
Xuange Zhang received the bachelor's degree in network engineering from Dalian Polytechnic University, China, in 2022. She is currently pursuing the master's degree in computer science with Beijing University of Technology, China. Her research interests include computer vision and intelligent surveillance understanding.



Jian Jin is with the Industrial Internet and IOT Research Institute, China Academy of Information and Communications Technology, Beijing, China. His research interests include large language models, blockchain, and distributed digital identifier.



Bo Liu received the M.S. degree in theoretical physics and the Ph.D. degree in computer science from Beijing University of Technology, Beijing, China. He is currently an Associate Professor with the School of Computer Science and Technology, Beijing University of Technology. His research interests include machine learning and computer vision.



Zhenzhen Jiao received the Ph.D. degree in computer science from the University of Chinese Academy of Sciences. Currently, he is the Director of the Teleinfo iF-Labs, China Academy of Information and Communications Technology. His research interests include blockchain, self-organized systems and networks, and computing power network.